

EMMANUEL MUWANGUZI

MSc Data Science | Data Analytics | Data Visualisation | Machine Learning

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<https://emmanuel-muwanguzi-portfolio.netlify.app>

PROFESSIONAL SUMMARY

MSc Data Science graduate with hands-on experience in data analysis, data visualisation, statistical analysis, machine learning and Python-based application development. Experienced in preparing and analysing datasets, identifying patterns and relationships, and translating complex data into clear, evidence-based insights. Analysed 9,994 product sales transactions using correlation analysis, segmentation and temporal analysis to investigate product performance, discount strategy and customer profitability. Developed an end-to-end web-based plant disease detection application using a lightweight Convolutional Neural Network (CNN), demonstrating experience in data preprocessing, model development, evaluation and deployment. Strong quantitative problem-solving, analytical reasoning and technical communication skills, with an interest in data engineering, data analytics and insight generation.

TECHNICAL SKILLS

Programming: Python, JavaScript, MATLAB, SQL

Data Analysis: Pandas, NumPy, Scikit-learn, Data Cleaning, Data Preprocessing, Data Transformation, Exploratory Data Analysis, Statistical Analysis, Correlation Analysis, Quantitative Analysis

Data Visualisation: Matplotlib, Seaborn, Google Colab, Data Visualisation, Statistical Visualisation, Data Storytelling, Trend Analysis, Pattern Analysis, Power BI

Machine Learning: Machine Learning, Deep Learning, Convolutional Neural Networks, Image Classification, Model Training, Model Evaluation, Computer Vision

Application Development: Flask, React, REST APIs, Frontend/Backend Integration, Machine Learning Model Deployment

Tools: TensorFlow/PyTorch, OpenCV, Git, GitHub, Jupyter Notebook, Google Colab, VS Code

EDUCATION

Roehampton University, London

MSc Data Science | 2025–2026

Master's Dissertation: Explainable Web-Based Tomato Disease Detection Using Lightweight CNN

Relevant coursework: Data Visualisation, Data Analysis, Machine Learning, Artificial Intelligence, Computer Vision, Deep Learning, Mathematics for Data Science

Makerere University, Kampala

BSc Computer Science | 2019–2023

Relevant coursework: Data and Information Management, Artificial Intelligence, Systems Analysis and Design

PROJECTS

Data Visualisation & Commercial Analytics

Roehampton University | 2026

Project: *A Data-Driven Exploration of Product Sales Performance, Discount Strategy, and Customer Profitability*

- Analysed **9,994 product sales transactions** to investigate product performance, discount strategy, customer profitability and regional and segment-level commercial trends.
- Cleaned, prepared and transformed transactional data using **Python and Google Colab**, creating an analysis-ready dataset for statistical analysis and visualisation.
- Applied **distributional analysis, correlation analysis, segmentation and temporal trend analysis** to identify relationships and performance patterns across products, customers, regions and time.
- Identified a strong negative relationship between discount rate and profit margin, with **Pearson correlations exceeding -0.75 across product categories ($p < 0.001$)**.
- Found that transactions with discounts above **30% were associated with near-zero or negative average profit margins**, highlighting the commercial impact of discount strategy.
- Analysed product sub-categories and identified **Tables, Bookcases and Supplies** as areas associated with cumulative losses over the analysis period.
- Examined customer and regional performance, identifying the **Consumer segment in the West and East regions** as significant contributors to absolute profit.
- Compared category performance and identified **Technology as the fastest-growing and most profitable category**, while Furniture presented greater profitability challenges.
- Identified a consistent **Q4 seasonal sales peak**, providing evidence relevant to inventory and capacity planning.
- Translated analytical findings into business-focused recommendations around discount controls, product portfolio strategy, resource allocation and seasonal planning.
- Communicated complex findings through clear static data visualisations, demonstrating the ability to transform transactional data into **evidence-based commercial insights**.

Technologies: Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab

Explainable Web-Based Tomato Disease Detection Using Lightweight CNN

MSc Project, Roehampton University | 2026

- Developed an end-to-end machine learning application for automated classification of tomato plant diseases from leaf images using a **lightweight Convolutional Neural Network (CNN)**.
- Prepared and preprocessed image data for machine learning, applying transformation techniques to support model training and evaluation.
- Developed and evaluated the CNN model using **Python and TensorFlow/PyTorch**, analysing model performance across image classification tasks.
- Built a **Flask backend** to process requests, connect application components and integrate the trained machine learning model for inference.
- Developed a **React frontend** enabling users to submit plant images and receive model predictions through a web-based interface.
- Integrated the trained model into the application and tested the complete workflow from image input and backend processing through model inference and presentation of prediction results.
- Investigated practical trade-offs associated with lightweight deep-learning models, including computational efficiency and suitability for application deployment.
- Applied structured experimentation and quantitative analysis to evaluate model performance and interpret results.
- Documented the methodology, implementation, experiments, results and limitations in the Master's dissertation.

Technologies: Python, TensorFlow/PyTorch, CNN, Machine Learning, Computer Vision, React, Flask, REST APIs, NumPy, Pandas

ADDITIONAL EXPERIENCE & CAPABILITIES

- **Data Analysis:** Data preparation, cleaning, transformation, exploratory analysis, statistical analysis, correlation analysis and interpretation of results.
- **Data Visualisation:** Translating quantitative analysis into clear visual representations of trends, relationships, distributions and comparisons.
- **Machine Learning:** Model development, training, evaluation, image classification and deep-learning workflows.
- **Software Development:** Integrating frontend, backend and machine-learning components into functional applications.
- **Problem Solving:** Problem formulation, experimental design, quantitative evaluation and structured investigation.
- **Technical Communication:** Technical documentation, analytical reporting and communication of complex findings to non-technical audiences.

- **Version Control:** Git and GitHub.